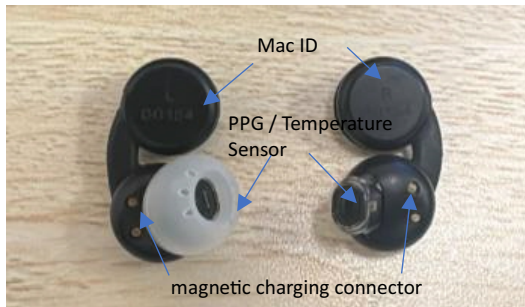


HLS Raw Data Version Manual

(Special FW for Raw data)

(FW Ver: PF2315 V0.1.0R0)

Headphones



Power On: Remove the earbud from the case to hear "Power on."

Power Off: Place the earbud back in the case.

Turn on the sensor: The earbuds have a wearing recognition function, and the sensors will automatically turn on when the earbuds are worn. Please refer to "How to Obtain Sensor Data via BLE" for PPG raw data.

Note:

Classic Bluetooth must be connected to the phone before using BLE data

Classic Bluetooth Name: PF2315-0XXX

BLE Name: PF2315-XXXL (Left side) or PF2315-XXXR (Right Side)

How to obtain sensor data via BLE

The earbuds use the BLE proprietary profile to transmit data, please use the following UUID to read the BLE information

Service UUID

40af0001-9479-43f6-ae95-c45f2afb9d2

Read Characteristic UUID

40af0003-9479-43f6-ae95-c45f2afb9d2

Write Characteristic UUID

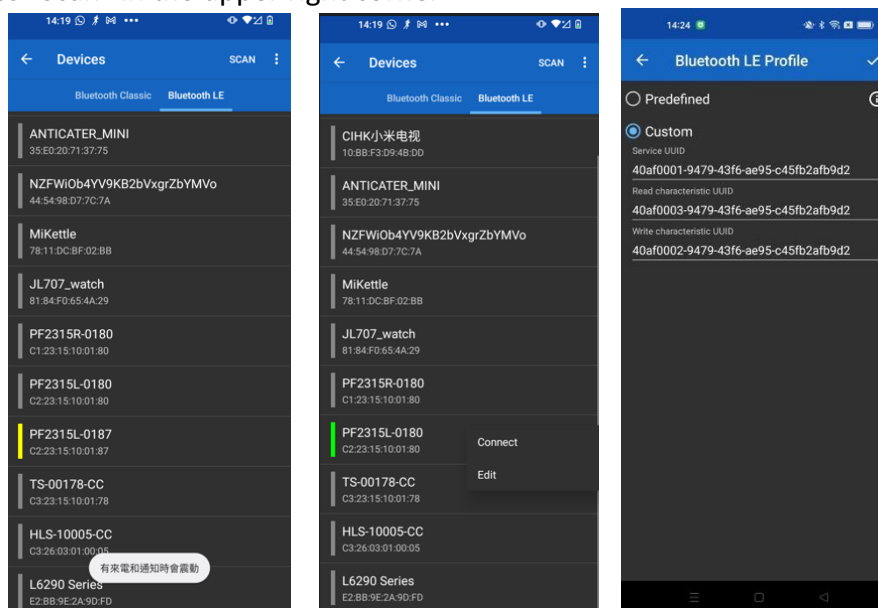
40af0002-9479-43f6-ae95-c45f2afb9d2

It is recommended to use the following Android APP for BLE communication

https://play.google.com/store/apps/details?id=de.kai_morich.serial_bluetooth_terminal

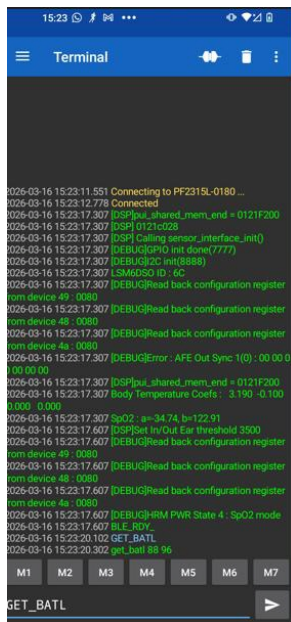
Step 1. Install and open the APP, and the relevant permissions must be turned on for the first time

Step 2. Press "Scan" in the upper right corner

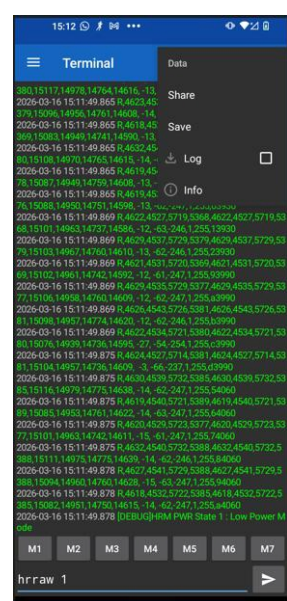
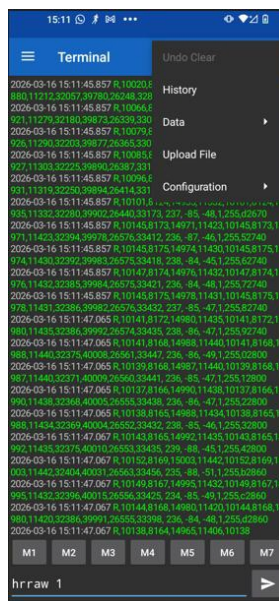
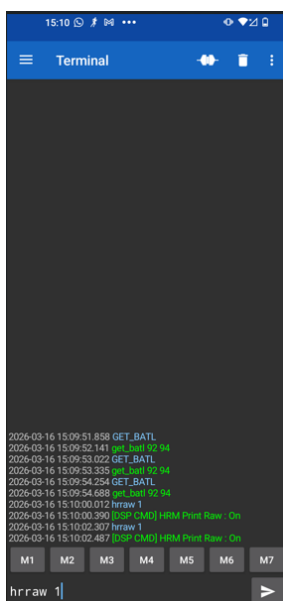
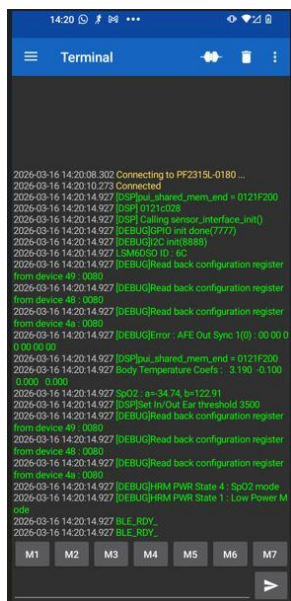


Step 3. Press and hold the appropriate earphone (the number of the earphone is the same as the number after the BLE name) and select "Edit" to set the UUID information.

Step 4. Go back to the search screen and tap the headset connection. After connecting, enter "GET_BATL" below and send. (To avoid earbud turning off)



Step 5. enter "hrraw 1" below and send.



Step 6. You can select log in the upper right corner to save the information.

15:20

Terminal

2026-03-16 15:20:36.057 R,9724,9104,12288,12266,9724,9104,12288,12266,25685,26991,24839,29205,-191,-110,-129,0,76,81630

2026-03-16 15:20:36.057 R,9728,9166,12288,12265,9728,9166,12288,12265,25690,26998,24844,29209,-191,-111,-129,0,76,a1630

2026-03-16 15:20:36.057 R,9729,9166,12290,12266,9729,9166,12290,12266,25689,26997,24844,29212,-192,-112,-129,0,76,b1630

2026-03-16 15:20:36.063 R,9729,9167,12291,12267,9729,9167,12291,12267,25688,26999,24843,29213,-191,-112,-129,0,76,c1630

2026-03-16 15:20:36.063 R,9729,9167,12291,12266,9729,9167,12291,12266,25691,26996,24846,29217,-192,-113,-129,0,76,d1630

2026-03-16 15:20:36.063 R,9733,9170,12300,12274,9733,9170,12300,12274,25694,27009,24851,29227,-193,-112,-129,0,76,41690

2026-03-16 15:20:36.063 R,9734,9174,12302,12278,9734,9174,12302,12278,25695,27009,24850,29225,-192,-111,-129,0,76,51690

2026-03-16 15:20:36.063 R,9734,9174,12303,12278,9734,9174,12303,12278,25694,27008,24852,29227,-192,-110,-130,0,76,61690

2026-03-16 15:20:36.208 R,9735,9172,12302,12278,9735,9172,12302,12278,25701,27014,24853,29232,-192,-111,-130,0,76,71690

2026-03-16 15:20:36.208

2026-03-16 15:20:36.208

2026-03-16 15:20:36.208

2026-03-16 15:20:36.208

M1

M2

1 to 4 columns PD data when the infrared LEDs are on

5 to 8 columns repeat of IR LEDs from column 1-4

9 to 12 column PD data when RED LED are on

13 to 15 column g-sensor 255 = 1G

16 – / 17 – Heart rate

18 – Internal Debug message

hrraw 1

PPG sensors



Data

The data of the headset is transmitted in ASCII mode and can be opened with a number of text editing tools.

Sampling Rate: 75 Hz

HRM Raw Data Print – Battery Usage

1. Without music: around 6 hours
2. With music at 50% volume: around 1.5 hours

Data Output:

- Raw PPG signal (Infrared and Red)
- Raw IMU signal (G-sensor)
- Heart Rate